



All about databases

Hello again. So far, you've seen how data can be gathered and analyzed to solve all kinds of problems. Next step, we're going to learn all about databases. As a refresher, a database is a collection of data stored in a computer system, but storage is just the beginning. You'll discover how databases make it possible to find the exact piece of information you need for your analysis. You'll also learn how to sort data in order to zoom in on what you need to generate insightful reports and much more. Then we'll go even deeper, and I mean really, really deep. I'm talking about metadata. You've probably heard someone say, Wow, that's so meta. Usually they're talking about something referencing back to itself or being completely self-aware. For example, if a character in a book knows she's in a book, that's meta. If you make a documentary about making documentaries, that's also meta. And here at Google, I constantly analyze how I analyze data. That's definitely meta.

I do that to give my work a quality check to make sure my methods are fair. And to be certain that I'm paying attention to any biases that might affect the outcome. As an analyst, you should do this too. Sometimes we get a little too close to our data. So stepping back and asking ourselves if our processes make sense is key. But let's back up just a bit and define metadata. Metadata is data about data. Like I said: deep.

Metadata is extremely important when working with databases. Think of it like a reference guide. Without the guide all you have is a bunch of data with no context explaining what it means. Metadata tells you where the data comes from, when and how it was created, and what it's all about.

Up next, you'll learn how to take data from a database or another source and bring it into a spreadsheet. You'll do this either by importing it directly or by using SQL to generate the request. And once you have data in a spreadsheet, the possibilities are endless. Everything we're about to cover is a very important part of the prepare phase of the data analysis process. It's how data analysts figure out which kind of data is going to be most helpful to them. If you have the right data, you're much more likely to be able to solve your business problems successfully. So, ready to tap into the incredible power of databases? Let's go!